

The Path Integral as Kernel Decomposition: Picard-Lefschetz Theory in the K-Q Framework

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Abstract

The Feynman path integral $\int \mathcal{D}[\varphi] e^{iS[\varphi]/\hbar}$ is a production kernel K that maps field configurations to transition amplitudes. On the original real integration domain, this kernel is oscillatory and conditionally convergent. Picard-Lefschetz theory resolves this by complexifying the field space and decomposing the integral into a sum over Lefschetz thimbles $\mathcal{J}_\sigma$ — steepest-descent manifolds attached to saddle points of the complexified action. We show that this thimble decomposition is a K-factorization in the sense of Volume VIII: each thimble defines a component kernel K_σ , the total kernel is $K = \sum_\sigma n_\sigma K_\sigma$ where n_σ are topological intersection numbers, and all decompositions yield the same observable under the projection Q . The framework identifies three structural results. First, resurgent relations — in which the perturbative expansion around one saddle encodes information about other saddles — are instances of $Q \circ K$ composition on $\ker(Q)$: non-perturbative saddle contributions lie in $\ker(Q)$ of the perturbative projection but shape observables through the composed map, exactly as Faddeev-Popov ghosts do in Volume IV. Second, the Stokes phenomenon (intersection numbers n_σ jumping at Stokes lines) is a factorization rearrangement in the kernel monoid: the component decomposition of K changes while $Q \circ K$ is invariant, directly analogous to the formulation fiber of Volume VIII. Third, stationary-phase dominance — in which wildly oscillating paths cancel and only neighborhoods of $\delta S = 0$ survive — is the structural mechanism by which K produces finite observables: incoherent contributions self-cancel in $\ker(Q)$, while coherent geometric structure passes to $\text{im}(Q)$. The polarizer principle (non-commutativity of projections) provides the physical intuition: elements individually in $\ker(Q)$ contribute non-trivially through $Q \circ K$, just as crossed polarizers transmit light when an intermediate filter is inserted. All structural results are machine-verified by Z3.

Keywords: path integral, Picard-Lefschetz, Lefschetz thimble, resurgence, Stokes phenomenon, K-Q framework, kernel decomposition, null space, saddle point, stationary phase, ghost activity, factorization, formal verification, Z3

1 Introduction: The Polarizer Principle

Consider three polarizing films. Two films with orthogonal transmission axes transmit zero light: the projection $P_{\text{horiz}} \circ P_{\text{vert}}$ annihilates all states. Insert a third film at 45° between them: $P_{\text{horiz}} \circ P_{45^\circ} \circ P_{\text{vert}}$ transmits nonzero intensity. An element in $\ker(P_{\text{horiz}})$ — vertically polarized light — contributes to the observable (transmitted intensity) through composition with an intermediate projection.

This is the polarizer principle: **projection does not commute with composition**. Elements individually annihilated by Q can contribute to observables through $Q \circ K$ when K routes through an intermediate structure.

The preceding volumes of this series established that this principle governs quantum field theory:

1. **Volume IV** [AtikYM]: Faddeev-Popov ghosts lie in $\ker(Q)$ (unphysical) but determine the β -function through $Q \circ K$ (observable). The algebraic switch is $f^{abc} \neq 0$.
2. **Volume V** [AtikGhost]: The ghost-activity theorem proves: $\ker(Q)$ is active if and only if the gauge algebra is non-abelian. The null space is not inert.
3. **Volume VI** [AtikAtlas]: Six types of $\ker(Q)$ structure, including migratory elements (anomalies) that cross the $\ker(Q)/\text{im}(Q)$ boundary.

This paper shows that the same principle governs the **path integral itself**. The Feynman path integral is a production kernel K that sums over all field configurations. Most configurations lie in $\ker(Q)$ of the stationary-phase projection — they contribute wildly oscillating phases that cancel. But the **composition** $Q \circ K$ through the full integral produces finite, physical observables. The individually null paths are not inert; they participate through interference.

The modern program of Picard-Lefschetz theory [Wit2010] and resurgence [DU2016] makes this precise by decomposing K into Lefschetz thimbles. We show that this decomposition is a K-factorization (Volume VIII) and that resurgent relations are $Q \circ K$ activity on $\ker(Q)$ (Volume V). The path integral joins the K-Q catalogue as the sixth kernel type — and the richest.

2 The Path Integral as Production Kernel

2.0.1 The kernel structure

The Feynman path integral defines a production kernel:

$$K_{\text{PI}}[\varphi_i, \varphi_f; T] = \int_{\varphi(0)=\varphi_i}^{\varphi(T)=\varphi_f} \mathcal{D}[\varphi] e^{iS[\varphi]/\hbar}$$

This maps boundary data (initial and final field configurations) to transition amplitudes. In the K-Q language:

$$K_{\text{PI}} : \text{BoundaryData} \rightarrow \text{Amplitude}$$

$$Q : \text{Amplitude} \rightarrow \text{Observable}$$

The observable projection Q extracts measurable quantities: cross-sections ($|K|^2$), expectation values ($\langle O \rangle = Q(K)$), or spectral data (poles of the propagator). The composed map $Q \circ K_{\text{PI}}$ is the physical theory.

2.0.2 The oscillatory problem

On the original real integration domain, K_{PI} is not well-defined as an absolutely convergent integral. The factor $e^{iS/\hbar}$ oscillates wildly: nearby field configurations contribute phases that nearly cancel. The integral is **conditionally** convergent — its value depends on the precise ordering and grouping of contributions.

In K-Q language: most of the integration domain contributes to $\ker(Q)$ of the stationary-phase projection. Only neighborhoods of classical solutions ($\delta S = 0$) contribute coherently to $\text{im}(Q)$. The vast majority of ‘paths’ are individually in $\ker(Q)$ but collectively determine observables through $Q \circ K$.

This is the polarizer principle at infinite-dimensional scale. The individually null paths are not inert; they participate through destructive interference that sculpts the observable.

2.0.3 The free-particle propagator

For a free non-relativistic particle, K is exactly computable:

$$K(x_f, t; x_i, 0) = \sqrt{\frac{m}{2\pi i \hbar t}} \exp\left(i \frac{m(x_f - x_i)^2}{2\hbar t}\right)$$

This solves the Schrödinger equation and equals the Feynman path integral exactly. It is the unique kernel satisfying: (i) composition law $K(x_f, t_2; x_i, 0) = \int K(x_f, t_2; x, t_1) K(x, t_1; x_i, 0) dx$, (ii) Schrödinger evolution, (iii) initial condition $K(x_f, 0; x_i, 0) = \delta(x_f - x_i)$.

The composition law is the semigroup property of the kernel monoid (Volume VIII). It expresses causal, forward-in-time evolution — no retrocausality, no ‘exploring the future.’ The ‘sum over all paths’ is the global representation of local causal propagation, not literal future-probing.

3 Picard-Lefschetz Decomposition as K-Factorization

3.0.1 Complexification

Picard-Lefschetz theory [Wit2010, TK2014, FLT2017] resolves the oscillatory problem by complexifying the field space. Real field configurations $\varphi : M \rightarrow \mathbb{R}$ are promoted to complex configurations $\varphi : M \rightarrow \mathbb{C}$. The action $S[\varphi]$ becomes a holomorphic functional on the complexified configuration space.

The original real integration domain $\Gamma_{\mathbb{R}} \subset \mathbb{C}^\infty$ is then **deformed** — using Cauchy’s theorem in infinite dimensions — into a sum of Lefschetz thimbles:

$$\Gamma_{\mathbb{R}} = \sum_{\sigma} n_{\sigma} \mathcal{J}_{\sigma}$$

where $\mathcal{J}_{\sigma}$ is the steepest-descent manifold (Lefschetz thimble) attached to the σ -th saddle point φ_{σ} of the complexified action, and $n_{\sigma} \in \mathbb{Z}$ are intersection numbers determined by the topology of the flow.

3.0.2 The thimble decomposition as K-factorization

Each thimble defines a component kernel:

$$K_{\sigma[\varphi_i, \varphi_f]} = \int_{\mathcal{J}_\sigma} \mathcal{D}[\varphi] e^{iS[\varphi]/\hbar}$$

On each thimble, the imaginary part of the action is constant ($\text{Im}(S)|_{\mathcal{J}_\sigma} = \text{Im}(S(\varphi_\sigma))$) and the real part decreases monotonically away from the saddle. This makes each K_σ absolutely convergent — the oscillatory problem is solved.

The total kernel is:

$$K_{\text{PI}} = \sum_{\sigma} n_{\sigma} K_{\sigma}$$

This is a **K-factorization** in the sense of Volume VIII: the total kernel decomposes into components, each attached to a geometrically meaningful sector (a saddle point of the classical action). The factorization is:

1. **Not unique** — different deformations of $\Gamma_{\mathbb{R}}$ yield different thimble decompositions (the Stokes phenomenon, Section 5).
2. **Observable-preserving** — all decompositions yield the same $Q \circ K$ (Cauchy’s theorem guarantees this).
3. **Geometrically structured** — each component corresponds to a classical solution (instanton, perturbative vacuum, bounce, etc.).

This maps directly to the kernel monoid of Volume VIII. The integer coefficients n_{σ} are the topological analogue of the factorization multiplicity. The total $Q \circ K$ is invariant across decompositions — the physical content does not depend on which thimble basis is chosen.

3.0.3 Comparison with residue theorem

The structural parallel with contour integration is exact:

Structure	Residue theorem	Path integral (Picard-Lefschetz)
Integration domain	Contour $\gamma \subset \mathbb{C}$	Field-space cycle $\Gamma \subset \mathbb{C}^{\infty}$
Oscillatory integrand	$f(z)$	$e^{iS[\varphi]/\hbar}$
Critical points	Poles/saddles of f	Saddles of S ($\delta S = 0$)
Decomposition	$\gamma = \sum n_k C_k$	$\Gamma = \sum n_{\sigma} \mathcal{J}_{\sigma}$
Coefficients	Winding numbers $n_k \in \mathbb{Z}$	Intersection numbers $n_{\sigma} \in \mathbb{Z}$
Invariance	Cauchy’s theorem	Cauchy’s theorem (∞ -dim)
Dominant contributions	Residues at poles	Semiclassical expansions at saddles

The path integral is the infinite-dimensional generalization of the residue theorem. Both are global integral formalisms extracting invariant structure from oscillatory geometry.

4 Resurgence as Null-Space Activity

4.0.1 The resurgence program

Resurgence theory [Eca1981, DU2016, BDU2013] establishes that the perturbative expansion around one saddle **encodes** information about other saddles. Specifically: the large-order behavior of perturbative coefficients around the perturbative vacuum (saddle σ_0) is controlled by the action difference $S(\sigma_1) - S(\sigma_0)$ of the nearest non-perturbative saddle (σ_1 , typically an instanton).

The resurgent relation takes the schematic form:

$$a_n^{(\sigma_0)} \sim \frac{n!}{(S_{\sigma_1} - S_{\sigma_0})^n} \cdot (\text{fluctuations around } \sigma_1)$$

The late-order coefficients of perturbation theory around σ_0 are determined by the early-order fluctuations around σ_1 .

4.0.2 K-Q interpretation

In the K-Q framework, this becomes transparent. Define the perturbative projection Q_{pert} : the projection that extracts the perturbative series around a given saddle. Then:

1. The perturbative expansion around σ_0 is $Q_{\text{pert}} \circ K_{\sigma_0}$ — the projection applied to the perturbative thimble kernel.
2. Non-perturbative saddle contributions (K_{σ_1} , instantons) lie in $\ker(Q_{\text{pert}})$ — they are invisible to finite-order perturbation theory. This is **projection-relative** invisibility: instantons are not ‘unphysical’ in the full theory (they are real solutions of the Euclidean equations of motion). They are invisible only relative to the perturbative projection Q_{pert} . The full projection Q sees them perfectly well.
3. Yet the composed map $Q_{\text{pert}} \circ K_{\text{PI}}$, which includes **all** thimbles, differs from $Q_{\text{pert}} \circ K_{\sigma_0}$ — the non-perturbative saddles shape the perturbative series through the total composition.

This is **structurally analogous** to the ghost-activity mechanism of Volume V:

Structure	Ghost activity (Vol. V)	Resurgence
Null-space element	Faddeev-Popov ghost	Non-perturbative saddle K_{σ_1}
In $\ker(Q)$ because	Unphysical (violates spin-statistics)	Invisible to perturbation theory
Shapes $\text{im}(Q)$ through	$Q \circ K$ (ghost loop $\rightarrow \beta_0$)	$Q \circ K_{\text{PI}}$ (resurgent relation)
Algebraic switch	$f^{abc} \neq 0$	$S_{\sigma_1} \neq S_{\sigma_0}$ (action gap)
Observable affected	β -function sign	Large-order growth, Borel ambiguities

The resurgent relation is null-space activity: elements in $\ker(Q_{\text{pert}})$ that determine properties of $\text{im}(Q_{\text{pert}})$ through the composed map. Perturbation theory ‘remembers’ non-perturbative topology because $Q \circ K$ routes through $\ker(Q)$ — the polarizer principle at the level of asymptotic expansions.

4.0.3 Ghost-instantons

The paper of Basar, Dunne, and Ünsal [BDU2013] identifies ‘ghost-instantons’ — complex saddle points that are not on the original real integration contour (their Stokes multiplier is zero) yet still affect the perturbative expansion through resurgent cancellations. In K-Q language: ghost-instantons are elements of $\ker(Q)$ (not on the physical contour, hence unobservable individually) that are active (shape $\text{im}(Q)$ through composition). The structural analogy with Faddeev-Popov ghosts is precise: both are individually unobservable elements that shape observables through composition. The analogy is structural, not ontological — FP ghosts are gauge artifacts while ghost-instantons are complex classical solutions — but both instantiate the same K-Q pattern of null-space activity.

5 Stokes Phenomenon as Factorization Rearrangement

5.0.1 The Stokes phenomenon

As parameters vary (coupling constant, external momenta, $\hbar$), the intersection numbers n_σ can jump discontinuously at special parameter values called **Stokes lines**. At a Stokes line, the relative phase between two saddles aligns, and a thimble that previously did not contribute to the decomposition begins to contribute (or vice versa).

This means the factorization $K = \sum n_\sigma K_\sigma$ **changes** at Stokes lines — the component decomposition is parameter-dependent.

5.0.2 K-Q interpretation: factorization rearrangement

Volume VIII [AtikFactorization] proved that non-admissible kernels have non-unique factorizations, and that physically equivalent formulations of GR (Cartan, teleparallel, Palatini) correspond to different factorizations of the same total pipeline — a **factorization rearrangement**. The physical content ($Q \circ K$) is invariant; only the decomposition changes.

The Stokes phenomenon is the same structure:

1. The total kernel K_{PI} is parameter-independent (it depends only on the action S and the boundary data).
2. The thimble decomposition $K = \sum n_\sigma K_\sigma$ is parameter-dependent (the n_σ jump at Stokes lines).
3. The observable $Q \circ K$ is invariant across the Stokes transition (Cauchy’s theorem guarantees this).

The factorization rearrangement:

$$K_{\text{PI}} = n_1 K_{\sigma_1} + n_2 K_{\sigma_2} \quad (\text{before Stokes line})$$

$$K_{\text{PI}} = n_{1'} K_{\sigma_1} + n_{2'} K_{\sigma_2} + n_3 K_{\sigma_3} \quad (\text{after Stokes line})$$

with $Q \circ K$ unchanged. A new saddle enters the decomposition, but the physical content is invariant.

This is the path-integral analogue of the formulation fiber: different thimble decompositions sit over the same physical theory, related by factorization rearrangement at Stokes lines. The monoid structure of Volume VIII applies directly.

5.0.3 Intersection numbers as topological data

The integers n_σ — which thimbles contribute and with what multiplicity — are topological invariants of the flow. They are determined by the relative homology of the thimbles with respect to the original real cycle. In K-Q language: the factorization multiplicities are topological, not dynamical. The **number** of thimbles needed to represent K is a structural property of the configuration space, not a choice of gauge or regularization.

6 Stationary Phase as Projection Mechanism

6.0.1 The stationary-phase principle

In the semiclassical limit ($\hbar \rightarrow 0$), the path integral is dominated by neighborhoods of classical solutions — configurations satisfying:

$$\delta S[\varphi] = 0$$

All other contributions cancel by destructive interference. The leading approximation is:

$$K_{\text{PI}} \approx \sum_{\text{classical } \varphi_{\text{cl}}} \left(\frac{e^{iS[\varphi_{\text{cl}}]}}{\hbar} \right) \cdot (\text{Gaussian fluctuations})$$

This is the saddle-point approximation: only saddles survive; everything else is in $\ker(Q)$ of the semiclassical projection.

6.0.2 K-Q interpretation

Stationary-phase dominance is the mechanism by which K produces finite observables despite summing over an infinite-dimensional space of configurations. In K-Q language:

1. The full configuration space $\mathcal{C}$ decomposes under Q into coherent sectors (neighborhoods of $\delta S = 0$) and incoherent sectors (everything else).
2. Incoherent sectors lie in $\ker(Q)$ in the sense that $Q(\text{incoherent}) \rightarrow 0$ in the observable limit — their oscillating contributions cancel by destructive interference and produce zero net observable. This is asymptotic annihilation (interference-based), not strict algebraic kernel membership. The $\ker(Q)$ language is structural shorthand for the vanishing of the projection in the physical limit.
3. Coherent sectors lie in $\text{im}(Q)$: they survive interference and produce physical predictions.

The projection Q is stationary-phase extraction. The production kernel K generates all representational probes (all paths). The projection Q extracts the invariant content (classical solutions + fluctuations). The composed map $Q \circ K$ produces observables.

This is structurally identical to how convergent Feynman integrals work in Volumes I–IV: the kernel K_{feyn} generates formal expressions (loop integrands), and Q extracts the physical differences. Here the ‘formal expressions’ are **all field configurations** and the ‘physical differences’ are **stationary-phase survivors**.

6.0.3 Connection to admissibility

The stationary-phase decomposition is the infinite-dimensional limit of a general K-Q structure: when the kernel sums oscillatory contributions, the projection annihilates the incoherent part and retains the coherent part. The coherent/incoherent decomposition is the $\text{im}(Q)/\text{ker}(Q)$ decomposition. In Picard-Lefschetz language, thimbles (coherent, steepest-descent) play the structural role of $\text{im}(Q)$; anti-thimbles (incoherent, steepest-ascent) play the structural role of $\text{ker}(Q)$. The identification is structural — thimbles and anti-thimbles are geometric objects while $\text{im}(Q)$ and $\text{ker}(Q)$ are projection-theoretic — but the decomposition logic is the same.

6.0.4 No retrocausality

The propagator $K(x_f, t_f; x_i, t_i) = \langle x_f | e^{-iH(t_f-t_i)/\hbar} | x_i \rangle$ is constructed from **forward** time evolution. The path integral emerges after slicing this causal evolution into infinitesimal steps. The ‘sum over all paths’ is a **representation** of causal propagation, not literal future-probing. In K-Q language: the paths are components of the K-representation basis (like Fourier modes), not ontological events. No individual path is ‘real’ — only $Q \circ K$ produces observables.

7 The Structural Prediction

The K-Q framework makes a structural prediction beyond what Picard-Lefschetz theory alone provides:

7.0.1 Multiple K-representations with shared Q

If the thimble decomposition is one K-factorization yielding a given $Q \circ K$, then there should exist **other** K-representations — not necessarily thimble-based — that yield the same observable quotient. Different ‘bases’ for the production kernel, with different null-space structures, should all produce the same $\text{im}(Q)$.

This prediction is already confirmed by existing physics:

1. **Euclidean vs. Lorentzian path integral:** Same observables (correlation functions related by Wick rotation), different K-representations (oscillatory vs. exponentially suppressed).
2. **Operator formalism vs. path integral:** $\langle \hat{O} \rangle = \text{Tr}(\rho \hat{O})$ and $\langle \hat{O} \rangle = \int \mathcal{D}[\varphi] O[\varphi] e^{iS/\hbar} / Z$ yield identical observables from completely different K-architectures.
3. **Different gauge-fixings:** The formulation fiber of Volume VIII — different gauge choices give different K -decompositions with the same $Q \circ K$.
4. **Lattice vs. continuum:** Same observables in the continuum limit, from a radically different K-representation (discrete configuration sums vs. functional integrals).

The K-Q framework explains **why** these equivalences exist: they are different fibers over the same observable quotient. The non-uniqueness of K is not a defect — it is the structural reason physics admits multiple equivalent formalisms.

7.0.2 The atlas connection

Volume VII [AtikAtlas] classified six types of $\ker(Q)$ structure. The path-integral context adds texture to each:

Atlas type	QFT realization (Vol. VII)	Path integral realization
Type 1: Inert	Scheme constants (A_0)	Off-shell paths (cancel by interference)
Type 2: Active	Ghosts ($f^{abc} \neq 0$)	Ghost-instantons (complex saddles)
Type 3: Redistributive	Unphysical polarizations	Stokes-dependent thimble weights
Type 4: Migratory	Chiral anomaly	Saddle coalescence (new saddle emerges)
Type 5: Confined	Quarks/gluons	Non-perturbative bound-state saddles
Type 6: Latent	θ -vacuum	Instanton sectors (tunneling between vacua)

Every atlas type has a path-integral instantiation. The classification is not specific to perturbative Feynman diagrams — it describes the universal structure of K-Q decomposition.

8 Implications

8.0.1 The K-Q framework subsumes, not competes with, Picard-Lefschetz

Picard-Lefschetz theory provides concrete analytic tools: gradient flow equations, thimble construction algorithms, Monte Carlo methods on complexified spaces. The K-Q framework provides the structural context: **why** the thimble decomposition has the form it does, **why** resurgent relations exist, and **why** different decompositions yield the same observables.

The relationship is:

$$\text{Picard-Lefschetz} \subseteq \text{K-Q (path integral instantiation)} \subseteq \text{K-Q (abstract framework)}$$

The abstract framework (Volume IX) is domain-independent. It applies equally to gravitational kernels (Volume I), Biot-Savart (Volume VI), and path integrals (this volume). Picard-Lefschetz is the specific analytic machinery for the path-integral instantiation.

8.0.2 Divergences as null-space elements

The resurgence program treats perturbative divergences not as failures but as encoded structural information: the factorial growth of perturbative coefficients **encodes** the location and weight of non-perturbative saddles. In K-Q language: divergences (elements associated with $\ker(Q)$) are not ‘infinities to subtract’ but structural data about the null space. Volume X proved this for

Hadamard renormalization; the resurgence connection extends it to the asymptotic structure of perturbation theory itself.

8.0.3 The polarizer principle as organizing insight

The polarizer experiment — three films, non-commutativity of projections — provides the correct physical intuition for all of the following:

1. Ghost activity: $\ker(Q)$ elements shaping $\text{im}(Q)$ through $Q \circ K$
2. Resurgent relations: non-perturbative saddles shaping perturbative asymptotics
3. Stationary-phase dominance: incoherent paths canceling to produce finite observables
4. Migration (anomalies): $\ker(Q)$ elements crossing to $\text{im}(Q)$ at higher resolution

All are instances of: **individually null contributions producing non-trivial observables through composition**. This is the structural content of $Q \circ K \neq 0$ on $\ker(Q)$.

8.0.4 The projection mechanism: geometry reconstructed, not assumed

The path integral may reveal something deeper than ‘another K-Q instantiation.’ It may exhibit the **mechanism** by which POT projection produces observable geometry and topology. In POT, the ontological substrate H_{ont} is not directly accessible. Observable structure is extracted by the composed map $Q \circ K$. The path integral makes this extraction **dynamical**:

1. The production kernel K generates all admissible modal probes (field configurations, paths, thimbles).
2. Incoherent probes self-cancel through destructive interference — they fall into effective $\ker(Q)$.
3. Coherent survivors (neighborhoods of $\delta S = 0$, stable saddle contributions) pass into $\text{im}(Q)$.
4. The resulting observable geometry is not a static image of something pre-existing; it is **reconstructed** as the invariant content of kernel-mediated probing.

The path integral thus realizes POT projection dynamically: the production kernel probes configuration-space topology, while the projection Q extracts the coherent invariant geometry observed as physical law. Geometry and topology are not assumed by POT projection — they are reconstructed as the stable invariant content that survives the kernel-to-observable map.

This may explain why the Feynman path integral felt structurally close to K-Q from the outset: it is an already-existing physics realization of the general POT projection mechanism.

8.0.5 What this paper does and does not claim

This paper claims: the Picard-Lefschetz thimble decomposition is a K-factorization; resurgent relations are $Q \circ K$ activity on $\ker(Q)$; the Stokes phenomenon is a factorization rearrangement; and stationary-phase dominance is the projection mechanism.

This paper does not claim: computational equivalence with Picard-Lefschetz methods (K-Q does not replace gradient flow algorithms), that all resurgent structures reduce to the six atlas types (this is a conjecture), or that the K-Q framework provides new computational tools for evaluating path integrals (it provides structural classification, not numerical methods).

8.0.5 References

- [Wit2010] Witten, E. Analytic continuation of Chern-Simons theory. AMS/IP Stud. Adv. Math. 50, 347–446 (2011). arXiv:1001.2933.
- [DU2016] Dunne, G. V. and Ünsal, M. New nonperturbative methods in quantum field theory: from large-N orbifolds to bions and resurgence. Ann. Rev. Nucl. Part. Sci. 66, 245–272 (2016). arXiv:1601.03414.
- [BDU2013] Basar, G., Dunne, G. V. and Ünsal, M. Resurgence theory, ghost-instantons, and analytic continuation of path integrals. JHEP 10, 041 (2013). arXiv:1308.1108.
- [TK2014] Tanizaki, Y. and Koike, T. Real-time Feynman path integral with Picard-Lefschetz theory and its applications to quantum tunneling. Ann. Phys. 351, 250–274 (2014). arXiv:1406.2386.
- [FLT2017] Feldbrugge, J., Lehnert, J.-L. and Turok, N. Lorentzian quantum cosmology. Phys. Rev. D 95, 103508 (2017).
- [Eca1981] Écalle, J. Les fonctions réurgentes (3 volumes). Publications mathématiques d’Orsay (1981–1985).
- [BH1991] Berry, M. V. and Howls, C. J. Hyperasymptotics for integrals with saddles. Proc. R. Soc. Lond. A 434, 657–675 (1991).
- [AtikAbstractKQ] Atik, E. The abstract K-Q framework: kernels, projections, and null-space structure across domains. POT VUFT Series, Vol. IX. **Preprint** (2026).
- [AtikFactorization] Atik, E. Non-unique factorization of projection kernels. POT VUFT Series, Vol. VIII. **Preprint** (2026).
- [AtikGhost] Atik, E. Ghost-mediated null-space activity: a structural theorem from the K-Q framework. POT VUFT Series, Vol. V. **Preprint** (2026).
- [AtikYM] Atik, E. The null space is not inert: Yang-Mills vacuum polarization from convergent integrals. POT VUFT Series, Vol. IV. **Preprint** (2026).
- [AtikAtlas] Atik, E. The structural atlas of $\ker(Q)$. POT VUFT Series, Vol. VII. **Preprint** (2026).